Dawid Ciupka

DEVOPS / PLATFORM ENGINEER

I build systems, automate everything, and make infrastructure actually work.

Interactive Portfolio & Projects   GitHub   dawidciupka@gmail.com   +48 794 219 389

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ABOUT

DevOps / Platform Engineer with a background in Industrial Automation. I build and operate real infrastructure — self-hosted Kubernetes clusters, CI/CD pipelines, IoT integrations, and embedded systems. I work at the intersection of systems engineering and software, bridging the gap between hardware and cloud-native tooling.

Previously a PLC programmer (Allen Bradley, Siemens) and mechanical designer (Autodesk Inventor), now focused on Kubernetes, Docker, Python, MQTT and making infrastructure actually work.

Automate first   Build real things   Hands-on engineering

PROFESSIONAL PROJECTS

Smart Building Platform

- **Context:** Client from the building management sector.
- **Scope:** Deployment and configuration on a self-hosted Kubernetes cluster managed via Rancher. Full platform architecture design, TLS certificate management, and enterprise authentication integration.
- **Stack:** Kubernetes · Rancher · Docker · SAML
- **Result:** Production-ready smart building platform with secure SSO authentication and containerized deployment.

Environment Monitoring System

- **Context:** Client from the pharmaceutical sector (confidential).
- **Scope:** Architecture design and deployment of an environment monitoring platform on Kubernetes. Handled full PKI and certificate lifecycle, implemented OAuth 2.0 and SAML-based SSO for enterprise identity provider integration.
- **Stack:** Kubernetes · Rancher · OAuth 2.0 · SAML
- **Result:** Secure, compliant monitoring platform integrated with client's enterprise identity infrastructure.

PERSONAL PROJECTS

Kubernetes Cluster (Rancher & HA)

- **Problem:** Need for a production-grade, highly available local environment to orchestrate microservices and IoT workloads without recurring cloud costs.
- **Solution:** Designed and deployed a multi-node K8s cluster using Rancher for management. Implemented MetalLB for Load Balancing, Longhorn for distributed block storage, and Traefik as Ingress Controller. Integrated Prometheus and Grafana for full-stack observability.
- **Stack:** Kubernetes · Rancher · Longhorn · MetalLB · Prometheus · Grafana · Docker
- **Grafana:** status.appevil.pl

Home Lab — Proxmox + GitOps (Flux)

- **Problem:** Wanted a reproducible, fully declarative home lab — from bare Proxmox nodes to running apps — that I can rebuild from scratch without manual clicking.
- **Solution:** PowerShell pipeline bootstraps a Proxmox cluster from Windows — idempotent SSH key and ssh-mesh distribution, host-key scanning, pvecm cluster creation and quorum handling with a clean reset path. On top, a Flux GitOps repo continuously reconciles the cluster from Git, with apps managed declaratively via Kustomize (web apps, Cloudflare tunnel, observability stack).
- **Stack:** Proxmox · PowerShell · Flux CD · Kustomize · Kubernetes · Git · Cloudflare Tunnel
- **Result:** A repeatable bare-metal-to-apps home lab: cluster bootstrap is automated and the running state is driven entirely from Git.

LifeOS — Self-Hosted Agentic LLM Platform (in progress)

- **Problem:** Wanted a private, self-hosted personal assistant for calendar, notes and reminders — driven by an LLM, without sending personal data to third-party cloud AI providers.
- **Solution:** Telegram assistant backed by a self-hosted LLM (Ollama, qwen3-coder) with multi-provider routing. LangGraph multi-agent workflows — a supervisor delegating to specialized agents — handle intent, ChromaDB + sentence-transformers provide semantic long-term memory (RAG), and a Django REST backend with Celery runs async tasks and Google Calendar sync.
- **Stack:** Python · Django · LangGraph · Ollama · ChromaDB · Celery · PostgreSQL · Redis · Docker · Kubernetes
- **Status:** Running on a self-hosted Kubernetes cluster (Rancher, Traefik) deployed via GitHub Actions CI/CD with a self-hosted runner. Agent workflows, semantic memory and calendar sync working; actively expanding agents and RAG.

Chessmaster — Chess Opening Trainer

- **Problem:** Most opening trainers reward rote memorization. I wanted a tool that teaches you to understand moves, with real engine feedback.
- **Solution:** Full-stack trainer where Stockfish scores every move (centipawn-loss classification), shows top continuations, idea arrows and an eval bar. Includes gambit training, a test mode with spaced repetition and opening unlocking, PGN game analysis, mistakes replayed as puzzles, themed tactics on the Lichess Puzzle DB (~3M positions), a position editor and JWT-based accounts.
- **Stack:** Next.js 15 · TypeScript · Tailwind · FastAPI · Stockfish 16 · python-chess · PostgreSQL · Docker · Kubernetes
- **Live:** chessmaster.appevil.pl

Hardware Random Generator (Teensy)

- **Problem:** Software PRNGs are not truly random — insufficient for cryptographic or security-sensitive use cases.
- **Solution:** Microcontroller-based entropy source using hardware noise. Data exposed via USB interface (migrating to Ethernet) and consumed by backend services.
- **Stack:** Teensy · C++ · Python · USB/Ethernet
- **Result:** True hardware entropy source integrated with software backend.

SKILLS

INFRASTRUCTURE
Kubernetes
Rancher
Docker
Terraform
Ansible
PowerShell

CI/CD & AUTOMATION
GitHub Actions
Bash scripting
Linux

DEVELOPMENT
Python
Django
REST API
PostgreSQL

NETWORKING
DNS, TCP/IP
Reverse Proxy
MQTT

EMBEDDED / IOT
Raspberry Pi
Teensy
MQTT

INDUSTRIAL
PLC (Allen Bradley)
Siemens TIA Portal
Autodesk Inventor

EXPERIENCE

DevOps / Platform Engineer
Oct 2024 – Present

Self-employed / Freelance

- Currently collaborating with an international company (remote, English-speaking team) — platform testing and validation on Kubernetes
- Deployed and configured applications on Kubernetes (Helm, manifests, TLS/ingress)
- Integrated applications with enterprise authentication (SSO, SAML, OAuth 2.0) and internal platforms via REST APIs
- Implemented CI/CD pipelines with GitHub Actions

Test Engineer (Embedded)
Jul 2023 – Oct 2024

Garmin

- Tested infotainment controllers — electronics (PCB) and firmware
- Owned test stations: repairs, diagnostics and process configuration

Industrial Automation Engineer

Feb 2019 – Jun 2023

3M

- Programmed PLCs: Allen Bradley (Studio 5000), Siemens (TIA Portal)
- Mechanical design and documentation in Autodesk Inventor
- Commissioning and troubleshooting of industrial automation systems
- HMI development and SCADA integration

LANGUAGES

Polish – Native

English – Professional